ZPL 1D & 2D Barcode Commands

Quick reference for Zebra Programming Language (ZPL)

1D (Linear) Barcodes

Barcode	ZPL Command	Typical use
Code 128	^BC	SKU, serial number, inventory
Code 39	^B3	Industrial / inventory
Code 93	^BA	Alphanumeric
UPC-A	^BU	Retail
UPC-E	^B9	Compact retail UPC
EAN-13	^BE	Retail / international
EAN-8	^B8	Small retail products
Interleaved 2 of 5 (ITF)	^B2	Numeric cartons / inventory
Industrial 2 of 5	^BI	Industrial applications
Standard 2 of 5	^BJ	Numeric applications
Codabar	^BK	Libraries / healthcare / logistics
LOGMARS	^BL	Military / logistics
MSI	^BM	Inventory / retail
Plessey	^BP	Legacy inventory
POSTNET	^BZ	Postal applications

2D Barcodes

Barcode	ZPL Command	Typical use
QR Code	^BQ	General-purpose, URLs, IDs, data
Data Matrix	^BX	Small labels, traceability, parts
PDF417	^B7	Documents, logistics, IDs
Aztec	^B0	Compact 2D data
MaxiCode	^BD	Parcel / shipping

Useful ZPL Examples

`Code 128 – ^BC`

```
^BCN,60,Y,N,N
^FDABC423030926001^FS
```

`QR Code – ^BQ`

```
^BQN,2,6
^FDQA,ABC423/030926/1^FS
```

```
Data Matrix — ^BX
```

```
^BXN,5,200
^FDABC423/030926/1^FS
```

```
PDF417 — ^B7
```

```
^B7N,5,5,8,8,N,N
^FDABC423/030926/1^FS
```

For SKU + Date + Serial Number: Code 128 (^BC) is a strong choice when you need a readable 1D barcode. For very small traceability labels, Data Matrix (^BX) is often useful because it can store data in a compact area.

Note: Exact parameters and supported symbologies can vary by Zebra printer model and firmware. Always test the generated ZPL on the target printer/scanner.